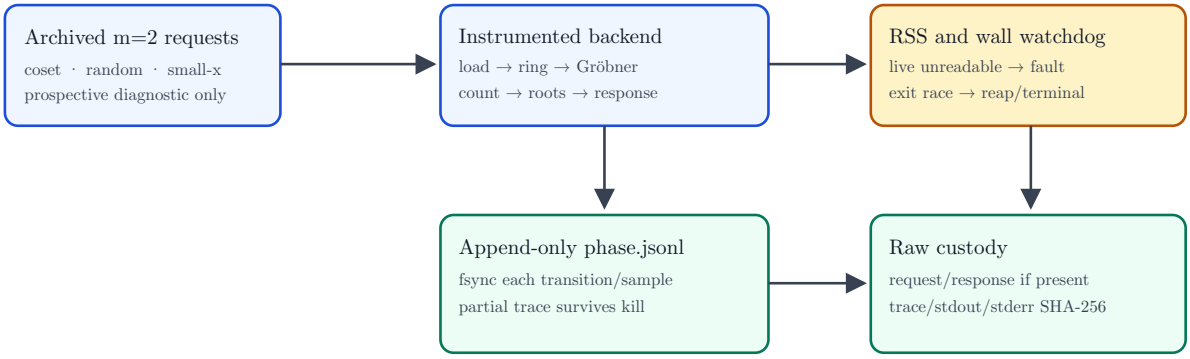
Prime63 monitor and phase instrumentation: Executor engineering report

TASK-20261018-3b41b1 · 2026-10-18 · implementation observation only. This work implements and tests process supervision and phase custody for a possible later positive-control diagnostic. It performs zero scientific solver calls and changes no hypothesis, experiment, run, or factor-base ranking. The diagram shows the intended data path. The raw engineering evidence is validation.json, with source and frozen-input SHA-256 values in source_manifest.json.

Why this was needed

The sealed RUN-ICPERF-7ed866 attempted seven of 27 cells. All six planted positive controls reached their 300-second caps without responses. The first natural coset m=2 cell then stopped on `rss_monitor_unavailable_for_live_child`; the checker rejected that missing-response state. The other 20 natural cells were unattempted. DEC-20261018-e92c77 authorized this separate nonexecution implementation. The problem record keeps monitor reliability and positive-control solver feasibility as distinct gaps.

Prime63 monitor and phase custody — engineering source, no solver trial in this task



Validated here: 10 synthetic tests, including 20 fast exits. Historical RUN-ICPERF-7ed866: 6 positive calls censored at 300 s.

Implementation

- `monitor.py` samples the live root and recursively discovered children using the pinned Sage Python `psutil` 7.2.2. It resolves a root exit/zombie race by rechecking the child lifecycle, uses `null` for unknown RSS, and emits a distinct `infrastructure_monitor_error` for an unreadable live child. It retains the 8 GiB proposed sampled tree-RSS cap as a later protocol parameter; this task's synthetic tests use smaller local caps when needed. When the root exits, the watchdog enumerates the old private process group before signaling surviving descendants; inability to verify or signal a remaining group becomes an infrastructure fault.
- `phase.py` appends JSONL events with monotonic and wall timestamps, stream sequence, phase name, source SHA-256 and fsync on each event. A truncated final record is preserved and flagged with its raw hash. `instrumented_backend.py` marks input load, polynomial-ring and ideal construction, Gröbner, model count, root enumeration and response write. It preserves the frozen cap-before-variety logic and requires a future approved-launch hash binding at the CLI. It was not invoked on a polynomial system here.
- `driver.py` runs synthetic phases only and writes custody containing exact trace, stdout and stderr hashes; a future scientific launcher must supply admission and input/response bindings. The prospective diagnostic design prelists the three m=2 archived positives and a monitor-fault null control for Coordinator review. It is unapproved and contains no trial result.

Synthetic validation

The exact command was `/Volumes/SSD990/cryptanalysis/third_party/sage-binary/sage -python research/icperf_prime63_monitor_phase_20261018/fault_tests.py --validation research/icperf_prime63_monitor_phase_20261018/validation.json`, run with host process-enumeration access. Its machine-readable record reports **10/10 passing synthetic cases** (suite wall time is recorded by nanosecond timestamps in the validation record). Cases cover normal exit, abrupt exit, a zombie at sampling, NoSuchProcess followed by reap, an injected unreadable live child, sampled RSS-cap termination, 20 repeated fast exit/reap calls, a reaped-root group-membership permission fault, a teardown fault with retained raw custody, and a forced-timeout phase dry-run. The timeout dry-run left a durable partial `groebner`-named **synthetic** phase trace outside the response; its stored SHA-256 matched the trace bytes. No Sage polynomial, Gröbner, relation, or ECDLP operation was run.

The first in-sandbox test attempt exposed a host permission boundary: macOS denied `psutil.children(recursive=True)` through `sysctl` with EPERM. That path correctly returned an infrastructure error with unknown RSS. The same synthetic suite passed under host process-enumeration access. This difference must be a later run-admission check; an environment without tree-read permission cannot support a qualified process-tree cost observation.

Scope and limits

The monitor measures **sampled** process-tree RSS. A short memory spike between samples can be missed; a child that deliberately leaves the private process group is outside this implementation's confinement assumption. An unparseable partial trace remains hash-bound but must be classified as incomplete telemetry. A failure of the trace storage itself is outside this in-process custody guarantee and requires the future external supervisor to preserve and classify the failure. The tests establish synthetic lifecycle behavior on this host, not solver feasibility or performance. The selected m=2 coset, random and small-x encodings have different archived input support, but input support does not establish observed Gröbner degree, sparsity during reduction, relation yield or runtime advantage. No end-to-end index-calculus cost or matched rho comparison is provided by this task.

The next scientific step requires a separate fresh-ID, Coordinator-approved diagnostic with independent polynomial and full-point checking and fixed limits. Even a checked m=2 positive result would leave natural relations, useful rank, linear algebra, target recovery, complete cold and online cost, and the m=2..8 comparison across prime, Koblitz and ordinary-binary curves open.